

Tata CLiQ

Database Management System

Chapter-1

Introduction:

Tata CLiQ is one of India's leading e-commerce platforms, owned by the Tata Group and launched in 2016. It offers a wide range of products, including fashion, electronics, beauty products, luxury brands, footwear, home appliances, accessories, and lifestyle essentials. The platform is known for providing genuine branded products, secure online payment options, fast delivery services, and a convenient shopping experience. Every day, Tata CLiQ handles thousands of customer interactions, product updates, orders, payments, and deliveries. Managing such a large volume of information requires a reliable and well-organized database system that can efficiently store, retrieve, and update data whenever required.

The Tata CLiQ database is designed to manage all the important information related to customers, products, sellers, orders, payments, inventory, and deliveries in a structured manner. It helps reduce data redundancy, maintain data accuracy and consistency, and improve the overall performance of the e-commerce platform. The database also supports secure storage of customer and payment information while enabling administrators to monitor inventory, process orders, generate reports, and analyze business performance. Overall, the Tata CLiQ database plays a vital role in ensuring smooth business operations, enhancing customer satisfaction, and supporting the continuous growth and success of the platform.

Objective:

The main objective of the Tata CLiQ database is to design and develop an efficient database system that stores, organizes, and manages all business-related information in a structured manner. The database helps maintain customer details, product information, seller records, orders, payments, inventory, and delivery details accurately. It ensures quick data retrieval, minimizes data redundancy, improves data consistency, and provides secure storage of information. The database also supports smooth business operations, enhances customer satisfaction, and helps management generate reports for effective decision-making and overall business growth.

Key Objectives:

- To store and manage customer information securely.
- To maintain detailed records of products, categories, and brands.
- To manage seller information and product listings efficiently.
- To process customer orders accurately and without delays.
- To record and manage payment transactions securely.
- To track inventory levels and update stock availability.
- To monitor the status of orders from purchase to delivery.
- To reduce data redundancy and maintain data consistency.
- To ensure data security, integrity, and reliability.
- To generate reports for sales, customers, products, and inventory analysis.
- To improve customer satisfaction through efficient database management.
- To support business growth and effective decision-making by providing accurate and organized data.

Problem Statement:

Tata CLiQ is a large e-commerce platform that manages thousands of customers, products, sellers, orders, payments, and deliveries every day. Handling this large amount of data without a proper database system can lead to data duplication, inaccurate records, delayed order processing, inventory mismatches, and security issues. These problems can reduce operational efficiency and negatively affect the customer experience.

The Tata CLiQ database is designed to address these challenges by providing a centralized and well-structured system for storing and managing business data. It ensures accurate record management, secure payment information, real-time inventory tracking, efficient order processing, and quick data retrieval. The database also reduces data redundancy, maintains data consistency, and supports better decision-making through reliable reports, helping the platform operate smoothly and efficiently.

Motivation:

Tata CLiQ was created to provide customers with a trusted, authentic, and seamless online shopping experience. Many e-commerce platforms faced issues such as counterfeit

products, delayed deliveries, and inconsistent customer service. Tata CLiQ aimed to solve these problems by offering genuine branded products backed by the trusted Tata Group.

The platform also focuses on integrating online and offline shopping through its **Phygital** model, allowing customers to enjoy the convenience of online shopping while benefiting from physical store support. Its motivation is to build customer trust, improve shopping convenience, and deliver a premium retail experience.

Stakeholders:

The stakeholders of the Tata CLiQ database are the individuals and organizations that use or benefit from the system.

- **Customers** – Browse products, place orders, make payments, and track deliveries.
- **Sellers/Vendors** – List products, manage inventory, and fulfill customer orders.
- **Administrators** – Manage customer data, products, sellers, orders, and overall database operations.
- **Warehouse Staff** – Update stock availability and manage inventory records.
- **Delivery Partners** – Access order and shipping details to ensure timely delivery.
- **Payment Service Providers** – Process and verify secure online payment transactions.
- **Management** – Use reports and analytics to monitor business performance and make strategic decisions.
- **Tata CLiQ (Organization)** – Uses the database to ensure efficient operations, improve customer satisfaction, and support business growth.

Business Requirements:

The Tata CLiQ database should support the core business operations of the e-commerce platform by efficiently managing customers, products, sellers, orders, payments, inventory, and deliveries. The system must provide accurate and secure data storage while ensuring fast access to information for both customers and administrators.

The database should allow customers to register, browse products, place orders, make secure payments, and track their order status. It should enable sellers to manage product listings and stock availability, while administrators can monitor all business activities, generate reports, and maintain data integrity. The system should also minimize data redundancy, ensure data consistency, support secure transactions, and provide reliable backup and recovery to ensure smooth and uninterrupted business operations.

Functional Requirements:

The Tata CLiQ database should provide all the necessary functions required to support the daily operations of the e-commerce platform. It must allow users to perform various activities efficiently while ensuring accurate data management.

- Customer registration and login.
- Store and manage customer information.
- Add, update, delete, and view product details.
- Manage product categories and brands.
- Maintain seller information and product listings.
- Place, update, and cancel customer orders.
- Process and record payment transactions.
- Track order and delivery status.
- Manage inventory and update stock availability.
- Generate reports for customers, products, orders, sales, and inventory.
- Provide secure access to users based on their roles.
- Ensure data accuracy, consistency, and efficient retrieval of information.

Non-Functional Requirements:

The non-functional requirements define the quality, performance, security, and reliability of the Tata CLiQ database system. These requirements ensure that the system operates efficiently and provides a smooth user experience.

- **Performance:** The database should process queries and transactions quickly with minimal response time.
- **Security:** Customer and payment information must be protected through secure authentication and authorization.
- **Reliability:** The system should provide accurate and consistent data with minimal downtime.
- **Scalability:** The database should be capable of handling increasing numbers of customers, products, and transactions as the business grows.
- **Availability:** The system should be available 24/7 to support continuous online shopping.

- **Data Integrity:** The database should maintain accurate, complete, and consistent data at all times.
- **Backup and Recovery:** Regular backups should be performed, and the system should support data recovery in case of failures.
- **Maintainability:** The database should be easy to maintain, update, and enhance when new features are introduced.
- **Usability:** The system should provide easy access to information for both customers and administrators.
- **Compatibility:** The database should work efficiently with the Tata CLiQ website, mobile application, and other integrated business systems.